Analyzing Crime Data in Washington DC

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AIT-580

Abstract

This research study investigates crime patterns in Washington DC by addressing three key questions. With increasing crime in cities, it is very important to understand and analyze crime data to introduce better policies to prevent crimes and ensure better safety to people of Washington DC. This study analyzed crime patterns using a comprehensive data analysis approach which involves cleaning the data, calculating the response time, and analyzing the dataset using Python, R and AWS tools. Bar plots were used to visualize geographic variation. R was then used to calculate distances to some landmarks like the White House and Lincoln Memorial using the Haversine formula, frequency of crime around these places has been visualized using bar plots. The method of crime was compared across offense type and region using bar plots and heatmaps. These showed that there was a great variation in the response times of districts and wards, that crimes are more frequent around landmarks, and that the methods of committing crimes vary greatly with regions. These findings can help law enforcement and policy makers with specific recommendations on the location based strategy of prevention and policing.

Introduction

Crime analysis plays an important role in helping to create effective law enforcement strategies and guides urban planning. In Washington DC-a city with very diverse neighborhoods and iconic landmarks-crime patterns will vary according to a host of different geographical, social, and environmental influences. Even though a great volume of research has been carried out into urban crime, little is known about how such specific factors as response times, proximity to landmarks, and variations in crime methods across offense and area types are huge factors in the dynamics of crime within this city. This study revolves around addressing three critical questions: (1) What is the average response time for crimes? How does it differ across different districts or wards? (2) Is proximity to certain landmarks associated with crime incidents? (3) How do methods of crime vary with various types of offenses and geographical regions?

Understanding these aspects is important for several reasons: response times highlight the differences in efficiency related to emergency services and point to areas that need infrastructure or resource improvements, the association between crime and proximity to landmarks will inform the strategic deployment of law enforcement resources and enlighten urban development policies to reduce hotspots of crime. It allows investigation of variations of offenses and geographic variations in crime for localized patterns, thereby informing the design of effective strategies for prevention and more effective targeting by police. The current research is intended to fill that knowledge gap with such information useful to policymakers and law enforcement in Washington DC.

Literature Review

Crime analysis has been very important for urban planners and law enforcement agencies. Research on response times stresses how important such times are in public safety, which also varies significantly according to area due to infrastructure and resource implications [1]. Knowledge of the disparities will help achieve better functions of law enforcement efficiency. The influence of proximity to landmarks on crime has been studied in various cities, and it indeed shows that crime rates are higher near iconic landmarks due to increased foot traffic and wealth concentrations [2]. In Washington DC, similar patterns have been observed, but further investigation is needed to clarify this relationship. It is however noted that crime methods relate to geographical regions. Therefore, socio-economic factors explain the occurrence of some type of crime in each location [3]. However not much consideration has been given on how such factors interface in given specific urban contexts, like that of Washington DC. This literature highlights important gaps in the understanding of how response times, proximity to landmarks, and geographical regions influence crime, particularly in Washington DC, which this research will seek to address.

Materials and Methods

This research seeks to focus on key aspects of the crime pattern in Washington DC through hypothesis-driven research. The paper studies variation in response times across districts/wards, the relationship between incidents of crime with distance from major landmarks, and changes in methods used across offense type and geographic area. In addressing these questions, my research was aimed to bring out patterns that can be used to inform appropriate strategies for crime prevention and resource allocation. To answer the research questions, data analysis was conducted using a combination of Python, R and AWS tools. To demonstrate my knowledge of SQL, I have created a database, table, uploaded the dataset into newly created table and ran a couple of SELECT statements to explore the dataset using MYSQL. Preprocessing, handling missing values, calculating response times, and statistical analysis for patterns and trends will be done using Python. Spatial analysis, like distances from crime locations to major landmarks such as the White House and Lincoln Memorial, was conducted using R. This shall give insight into how proximity to high-profile areas influences frequencies. AWS was used to analyze how different methods of crime vary between offense types and across different geographical areas. Using data hosted in AWS S3, Sagemaker studio, Python was utilized for analysis to group data by offense type, then by crime method, and into regions. This approach allowed convenient handling of large datasets and offered the possibility of creating visualizations, such as bar plots and heatmaps, to show variation in regional and categorical methods of committing the crime.

Visualization is provided using Seaborn, Matplotlib, and ggplot2 in illustrating the findings. These visualization tools helped create visualizations ranging from bar plots to heatmaps and spatial maps to show disparities in response times, variations in methods of criminal activities, and landmark-related crime patterns. The current research will integrate advanced data analysis methods with effective visualization to provide actionable insight for law enforcement agencies and policy makers in Washington DC.

Results

The project was analyzed crime incident data with three primary hypotheses:

1.Crime response times vary significantly across different districts and wards.

To answer the above question, I have used python.

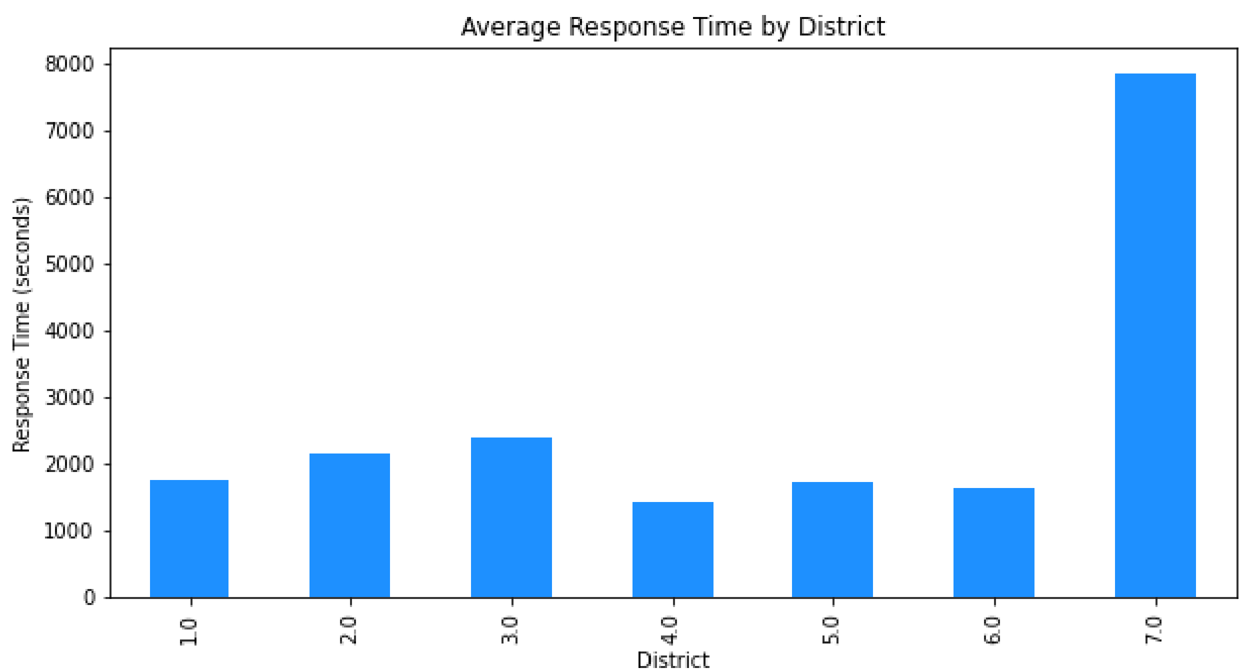


Fig. 1

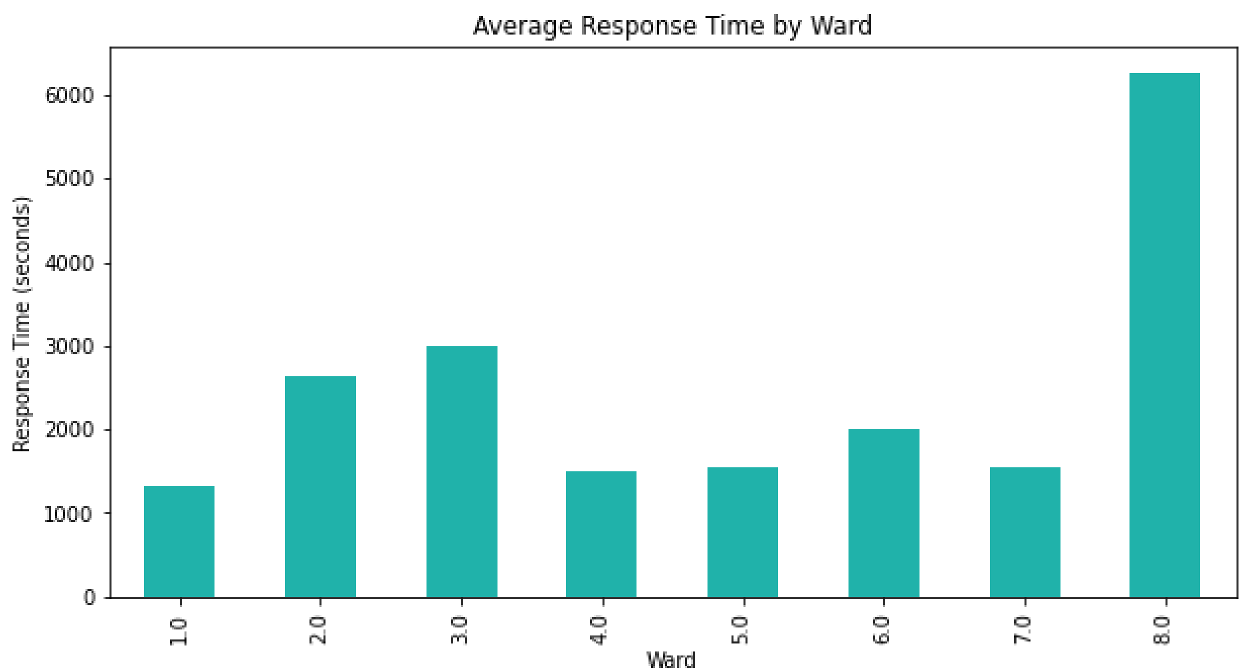


Fig. 2

The average response time for all incidents was about 2834 seconds at the ward level and about 2697 seconds at the district level. Based on the analysis at district-level, District 1 had the longest time to respond to all calls, which could point toward resource allocation issues, while District 4 had the quickest response times, reflecting better operational efficiency (Fig. 1). Similarly, at the level of wards, Ward 2 showed more efficiency in terms of response times relative to other wards (Fig. 2).

2.Is proximity to certain landmarks associated with crime incidents?

To answer this question, I have used R.

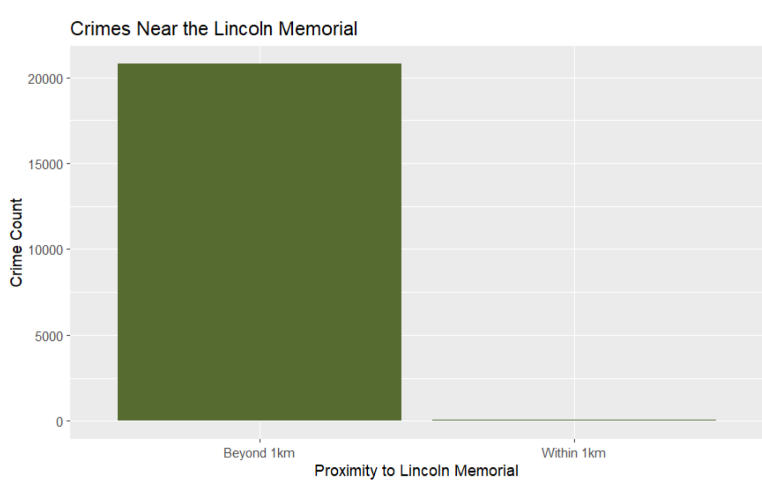


Fig. 3

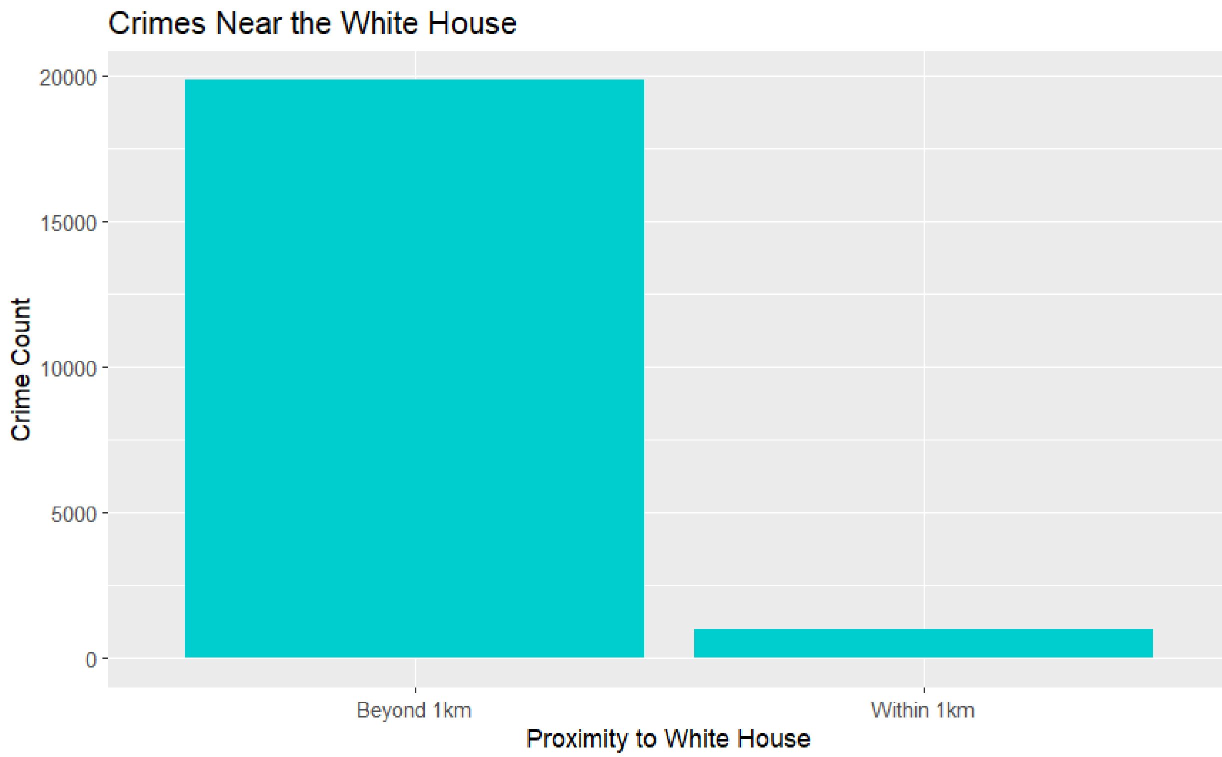
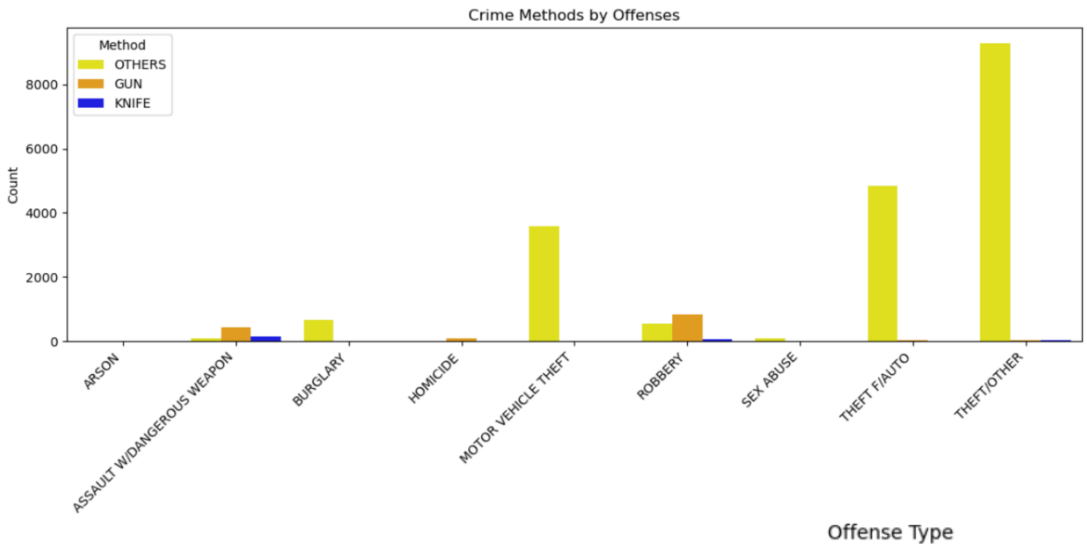


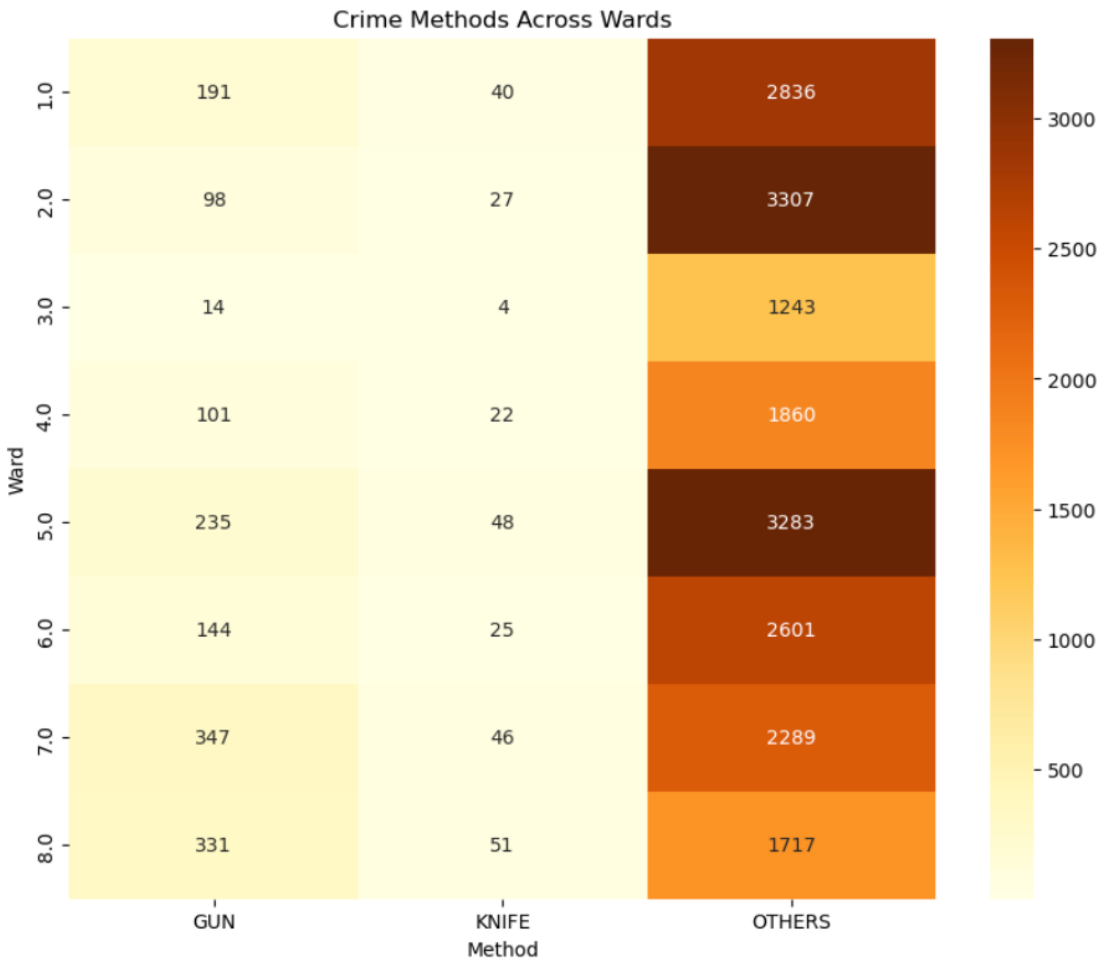
Fig. 4

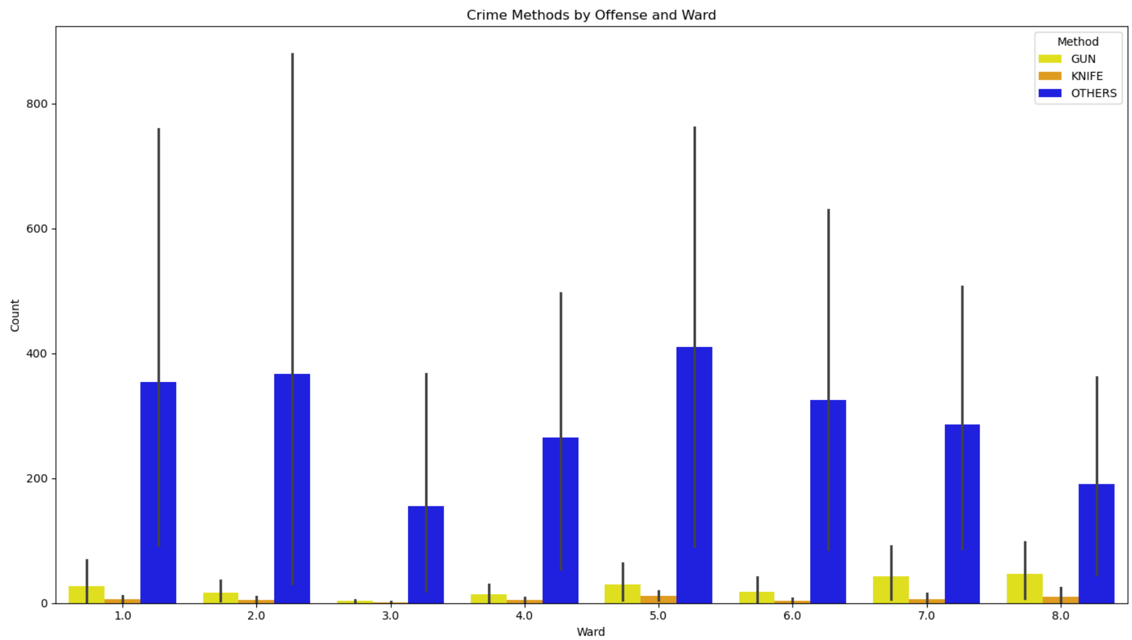
We can clearly see that there were more crimes near the White House than near the Lincoln Memorial within a 1 km radius, which might be indicative of the possible influence of pedestrian density and intensity of urban activities on crime frequency.

3) How do methods of crime vary with various types of offenses and geographical regions?

To answer this question, I have used AWS where my data set was residing in S3 bucket which I accessed using AWS Sagemaker, in there I created a jupyter studio to analyze and visualize the graphs using python.


Crime Methods by Offenses
Method
OTHERS
GUN
KNIFE
Count
0
2000
4000
6000
8000
ARSON
ASSAULT W/DANGEROUS WEAPON
BURGLARY
HOMICIDE
MOTOR VEHICLE THEFT
ROBBERY
SEX ABUSE
THEFT F/AUTO
THEFT/OTHER
Offense Type


Crime Methods Across Wards
Ward
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
191 40 2836
98 27 3307
14 4 1243
101 22 1860
235 48 3283
144 25 2601
347 46 2289
331 51 1717
GUN
KNIFE
OTHERS
Method
3000
2500
2000
1500
1000
500



Different offenses and different regions greatly vary in terms of crime methods. While the assaultive offenses were more related to firearms, theft-related offenses had been more opportunistic. Commercial districts present more opportunistic crimes, while residential areas have burglaries and assaults. This was further made clearer through visualizations that pointed out regional patterns, thereby making a call for tailored strategies in law enforcement. Being able to understand this variation therefore presents an avenue towards effective resource allocation and implementation of region-specific crime prevention measures.

Univariate and Multivariate Analyses of Each Type of NOIR Data in the Crime Dataset

Nominal Data:

OFFENSE

METHOD

WARD

NEIGHBORHOOD_CLUSTER

VOTING_PRECINCT

Ordinal Data:

WARD

PRIORITY_LEVEL

Interval Data:

REPORT_DAT

START_DATE

END_DATE

Ratio Data:

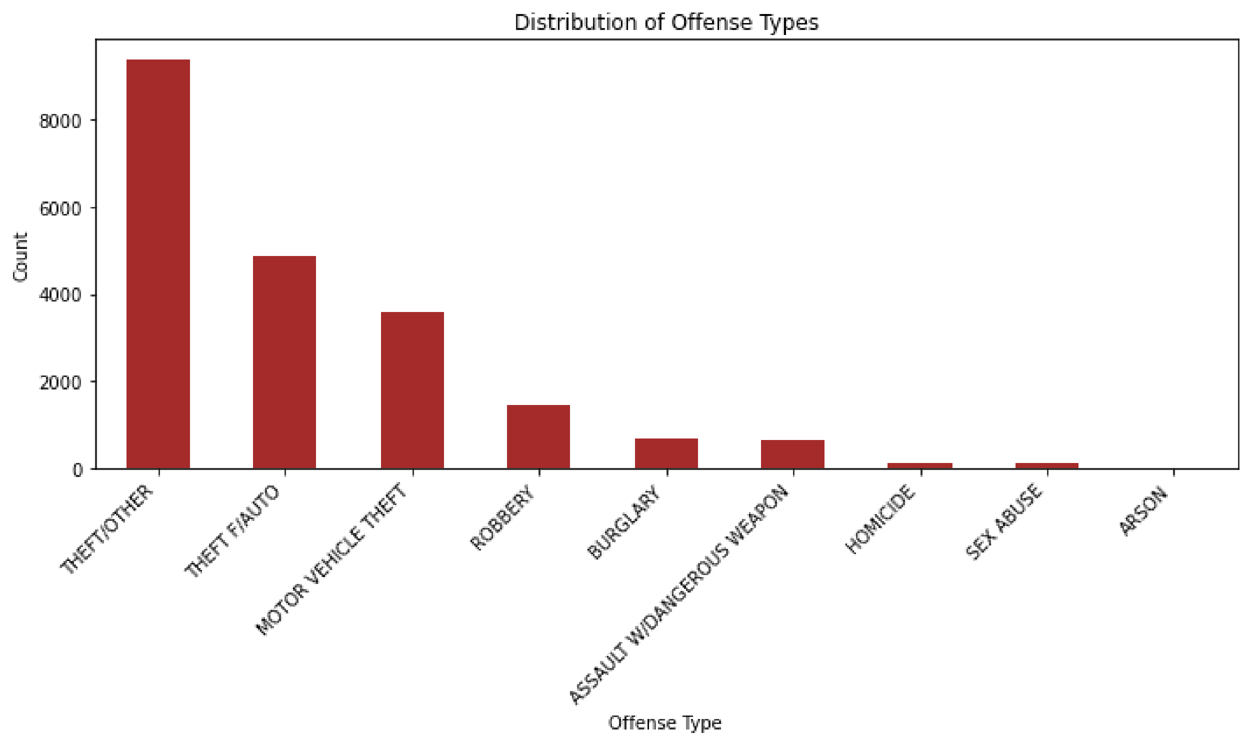
RESPONSE_TIME_MINUTES

LATITUDE

LONGITUDE

DISTANCE_TO_LANDMARK

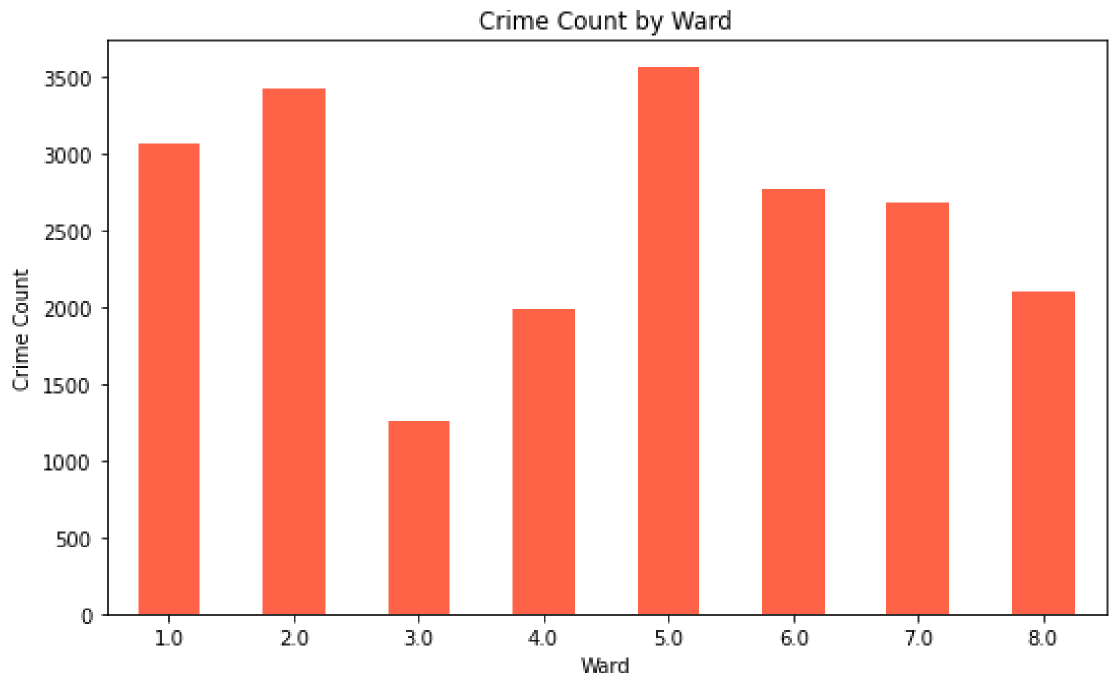
Univariate Analysis:
Nominal Data:



High counts for offenses such as motor vehicle theft indicate that these offenses are the highest in the dataset, while lower counts for offenses such as homicide show these are not as frequent.

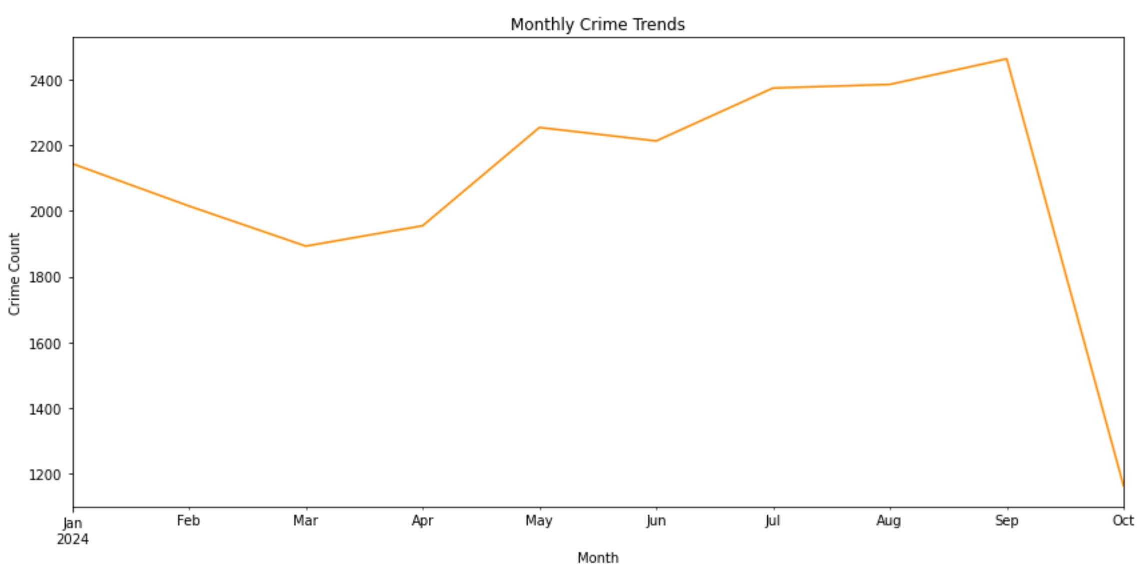
This distribution provides good insight into resource allocation, where law enforcement can prioritize strategies to address the most frequent offense types.

Ordinal:



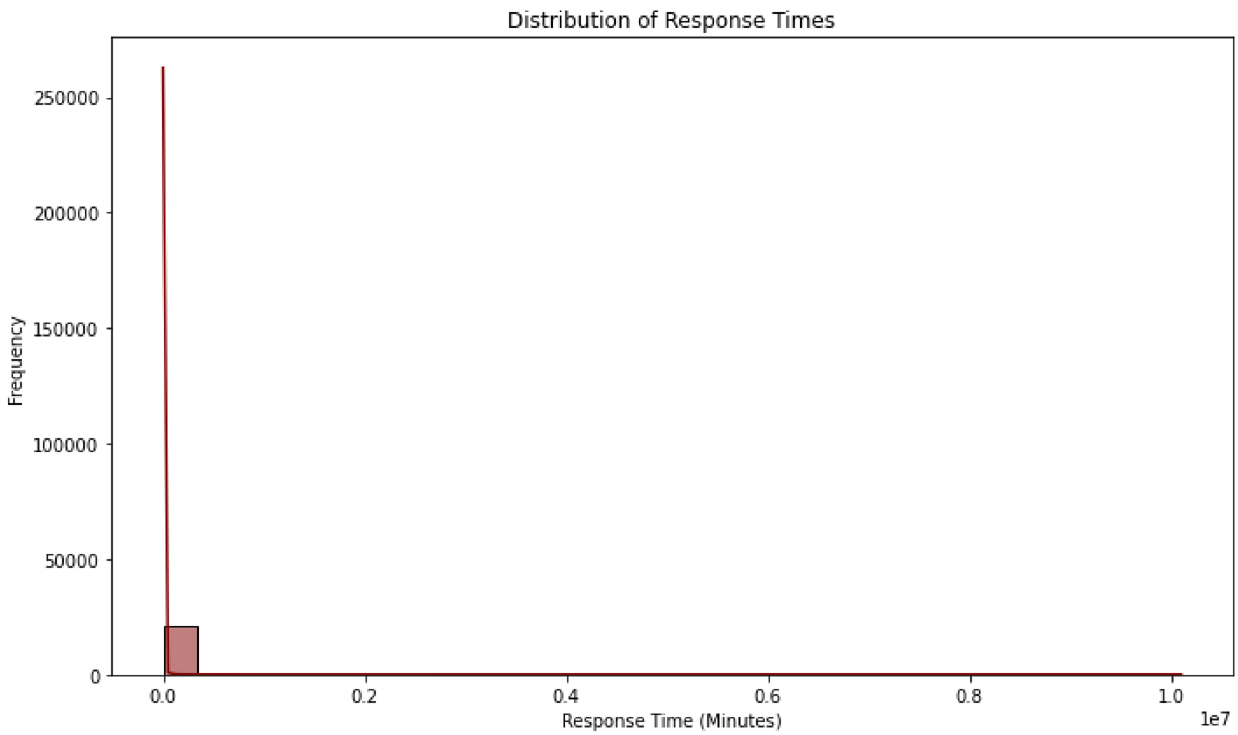
A high crime count bar in Wards 1 and 2 probably points to denser activities of crime due to socioeconomic factors or population density. Lower crime counts bars indicate Wards 7 and 8 have fewer reported crimes, which could either be from a low rate of crime flow or underreporting. This will provide an understanding whereby areas that require more policing and community support will be spotted.

Interval:



Peaks in any month might coincide with events or seasonal factors. For instance, spikes in the summer months might indicate higher levels of activity in better weather. When such patterns are identified, then law enforcement can prepare for predictable surges in crime.

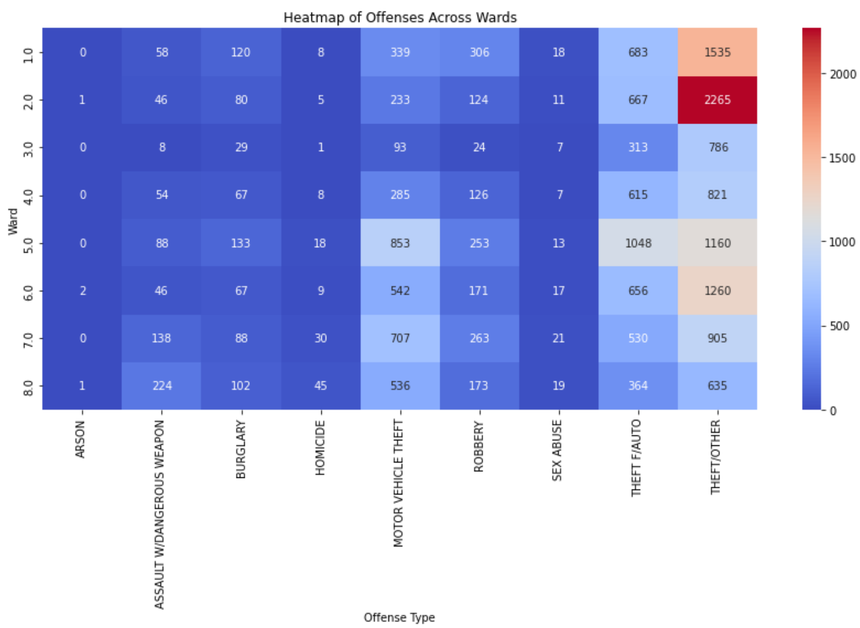
Ratio:



A concentration of response times around a specific value suggests average service efficiency. Extended tails indicate outliers, which may reflect delays due to infrastructure challenges. This analysis highlights areas needing operational improvements.

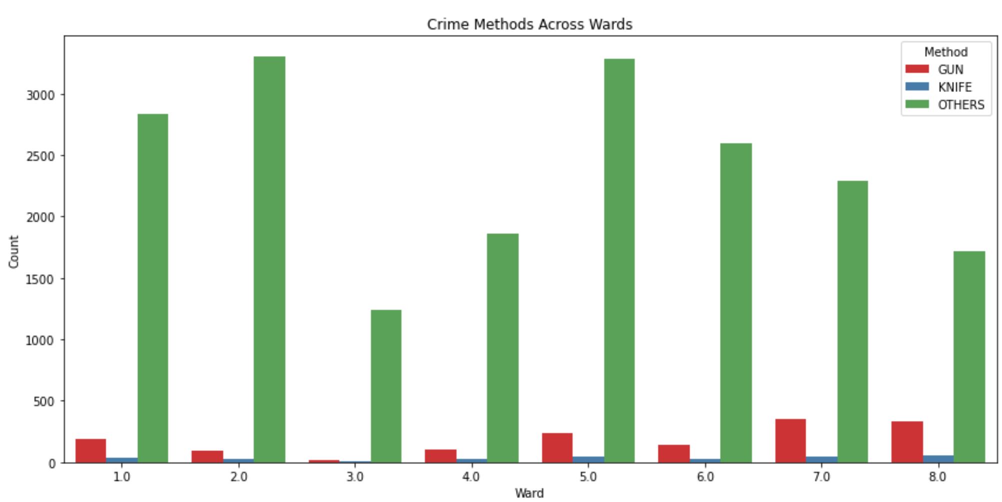
Multivariate Analysis

Nominal:



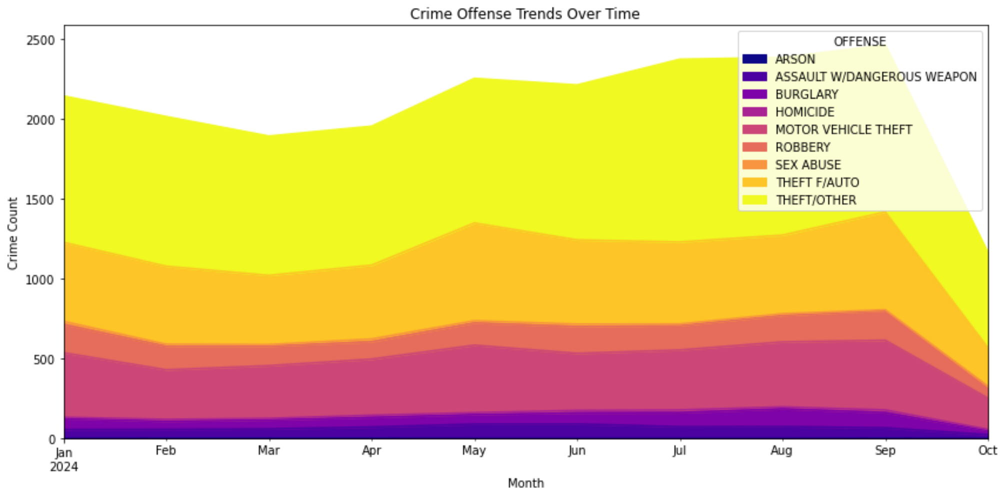
The colors change from blue to red for higher frequencies, pinpointing hotspots for certain infractions in these words. For example, although Ward 8 might report more assaults, Ward 2 has a high prevalence of larceny. This trend enables law enforcement to apply focused interventions in locations where certain crimes are more likely to occur.

Ordinal:



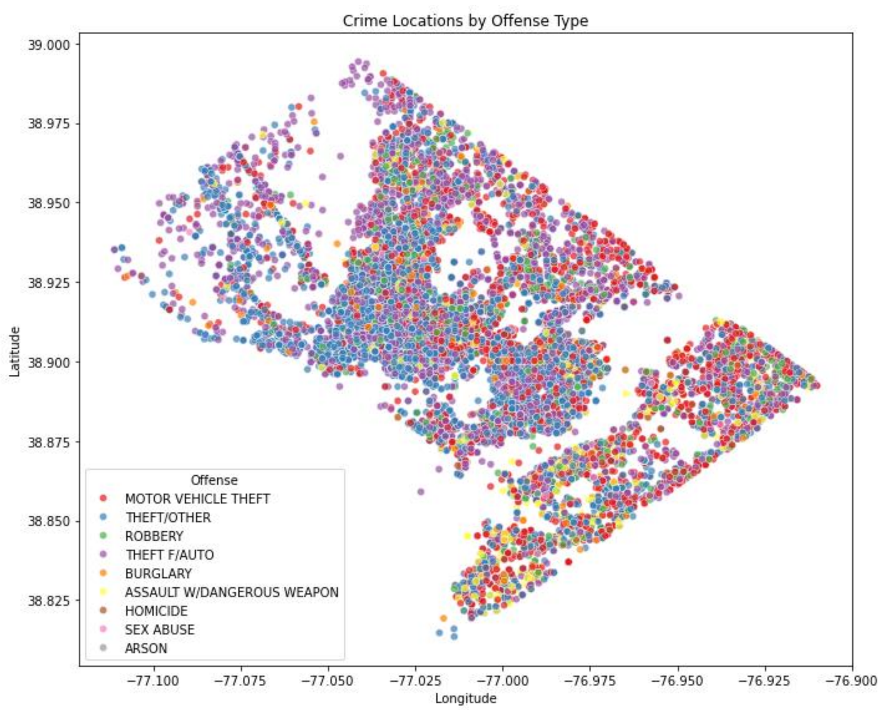
Some of the wards may have higher firearm crimes, while others report a higher number of unarmed crimes. For example, it could be observed that in Ward 3, certain methods have a high frequency and thus require community awareness programs. This insight would inform the deployment of ward-specific strategies to reduce types of crime.

Interval:



Gradual increases in theft offenses may point to economic factors. Sudden spikes in assault offenses might correlate to specific events. Indeed, this form of analysis helps predict future crime patterns and aids in the allocation of resources.

Ratio:



Densely populated clusters will show high-crime locations or commercial zones, while sparse points could be isolated incidences in residential areas. This will make it easier to find hotspots and to deploy resources strategically in accordance with them.

SQL

```
Enter password: ********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 9.0.1 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database ait_project;
Query OK, 1 row affected (0.02 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| ait580             |
| ait_project        |
| cs_project         |
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
| world              |
+--------------------+
9 rows in set (0.00 sec)

mysql> use ait_project;
Database changed
mysql> |
```

I have created a new schema named ‘ait_project’.

```
mysql> use ait_project;
Database changed
mysql> CREATE TABLE crime_data (
    ->     X DECIMAL(10, 6),
    ->     Y DECIMAL(10, 6),
    ->     CCN BIGINT PRIMARY KEY,
    ->     REPORT_DAT TIMESTAMP,
    ->     SHIFT VARCHAR(50),
    ->     METHOD VARCHAR(50),
    ->     OFFENSE VARCHAR(100),
    ->     BLOCK VARCHAR(100),
    ->     XBLOCK DECIMAL(10, 6),
    ->     YBLOCK DECIMAL(10, 6),
    ->     WARD VARCHAR(50),
    ->     ANC VARCHAR(50),
    ->     DISTRICT VARCHAR(50),
    ->     PSA INT,
    ->     NEIGHBORHOOD_CLUSTER VARCHAR(100),
    ->     BLOCK_GROUP VARCHAR(100),
    ->     CENSUS_TRACT VARCHAR(100),
    ->     VOTING_PRECINCT VARCHAR(100),
    ->     LATITUDE DECIMAL(9, 6),
    ->     LONGITUDE DECIMAL(9, 6),
    ->     BID VARCHAR(50),
    ->     START_DATE TIMESTAMP,
    ->     END_DATE TIMESTAMP,
    ->     OBJECTID BIGINT,
    ->     RESPONSE_TIME_MINUTES INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+-----------------------+
| Tables_in_ait_project |
+-----------------------+
| crime_data            |
+-----------------------+
1 row in set (0.00 sec)

mysql> |
```

Created a new table for my dataset, named-‘crime_data’.

```
mysql> load data local infile 'C:/Users/phani/Desktop/AIT 580/crime_data.csv' into table crime_data fields terminated by ',' lines terminated by '\n' ignore
 1 rows;
Query OK, 20860 rows affected, 368 warnings (0.36 sec)
Records: 20860  Deleted: 0  Skipped: 0  Warnings: 368

mysql> |
```

Uploaded the dataset file to the table using load infile command.

```
MySQL 9.0 Command Line Cli

mysql> select * from crime_data limit 10;
+--------+--------+----------+---------------------+----------+--------+---------------------+-------------------------------------------+--------+--------+-
-----+------+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+-----------+--------------------
-+---------------------+-----------+----------------------+
| X      | Y      | CCN      | REPORT_DAT          | SHIFT    | METHOD | OFFENSE             | BLOCK                                     | XBLOCK | YBLOCK |
WARD | ANC  | DISTRICT | PSA | NEIGHBORHOOD_CLUSTER | BLOCK_GROUP | CENSUS_TRACT | VOTING_PRECINCT | LATITUDE | LONGITUDE | BID       | START_DATE
 | END_DATE            | OBJECTID  | RESPONSE_TIME_MINUTES |
+--------+--------+----------+---------------------+----------+--------+---------------------+-------------------------------------------+--------+--------+-
-----+------+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+-----------+--------------------
-+---------------------+-----------+----------------------+
| 396920 | 138793 | 24091506 | 2024-06-16 05:01:01 | MIDNIGHT | OTHERS | MOTOR VEHICLE THEFT | 1500 - 1599 BLOCK OF U STREET NW          | 396920 | 138793 |
   1 | 1B   |        3 |  301 | Cluster 6            | 004300 3    |         4300 | Precinct 22     |   38.917 |  -77.0355 |           | 2024-06-16 03:00:00
 | 2024-06-16 04:48:00 | 606076223 |                   108 |
| 397792 | 134654 | 24077884 | 2024-05-24 02:16:30 | MIDNIGHT | OTHERS | MOTOR VEHICLE THEFT | 50 - 99 BLOCK OF DISTRICT SQUARE SW       | 397792 | 134654 |
   6 | 6D   |        1 |  103 | Cluster 9            | 010202 5    |        10202 | Precinct 142    |  38.8797 |  -77.0255 | SOUTHWEST | 2024-05-23 22:19:00
 | 2024-05-23 23:42:00 | 606076245 |                    83 |
| 396519 | 142279 | 24062904 | 2024-04-27 12:19:40 | DAY      | OTHERS | MOTOR VEHICLE THEFT | 4700 - 4799 BLOCK OF BLAGDEN TERRACE NW   | 396519 | 142279 |
   4 | 4E   |        4 |  404 | Cluster 18           | 002600 2    |         2600 | Precinct 48     |  38.9484 |  -77.0402 |           | 2024-04-27 11:26:00
 | 2024-04-27 11:30:00 | 606076267 |                     4 |
| 397824 | 141377 | 24106105 | 2024-07-11 10:17:39 | DAY      | OTHERS | THEFT/OTHER         | 4000 - 4099 BLOCK OF GEORGIA AVENUE NW    | 397824 | 141377 |
   4 | 4C   |        4 |  404 | Cluster 18           | 002503 1    |         2503 | Precinct 47     |  38.9403 |  -77.0251 |           | 2024-07-11 09:27:00
 | 2024-07-11 09:27:00 | 606076268 |                     0 |
| 398477 | 137709 | 24112374 | 2024-07-22 19:39:54 | EVENING  | OTHERS | ROBBERY             | 400 - 499 BLOCK OF N STREET NW            | 398477 | 137709 |
   2 | 2G   |        3 |  308 | Cluster 7            | 004802 1    |         4802 | Precinct 18     |  38.9072 |  -77.0176 |           | 2024-07-22 13:57:00
 | 2024-07-22 14:00:00 | 606076269 |                     3 |
| 403776 | 139530 | 24051500 | 2024-04-06 19:33:03 | EVENING  | OTHERS | THEFT/OTHER         | 2470 - 2599 BLOCK OF BALDWIN CRESCENT NE  | 403776 | 139530 |
   5 | 5C   |        5 |  503 | Cluster 24           | 009000 2    |         9000 | Precinct 139    |  38.9236 |  -76.9565 |           | 2024-04-06 16:30:00
 | 2024-04-06 16:35:00 | 606076272 |                     5 |
| 399887 | 136928 | 24017682 | 2024-02-04 06:17:26 | MIDNIGHT | OTHERS | THEFT/OTHER         | 300 - 399 BLOCK OF H STREET NE            | 399887 | 136928 |
   6 | 6C   |        1 |  104 | Cluster 25           | 008301 1    |         8301 | Precinct 83     |  38.9002 |  -77.0013 |           | 2024-02-04 02:24:00
 | 2024-02-04 03:22:00 | 606076280 |                    58 |
| 401095 | 140318 | 24019021 | 2024-02-06 17:20:12 | EVENING  | OTHERS | MOTOR VEHICLE THEFT | 1300 - 1399 BLOCK OF KEARNY STREET NE     | 401095 | 140318 |
   5 | 5B   |        5 |  504 | Cluster 22           | 009301 4    |         9301 | Precinct 73     |  38.9307 |  -76.9874 |           | 2024-02-06 06:30:00
 | 2024-02-06 17:15:00 | 606076281 |                   645 |
| 400798 | 140980 | 24019825 | 2024-02-08 04:18:16 | MIDNIGHT | OTHERS | THEFT/OTHER         | 3800 - 3899 BLOCK OF 12TH STREET NE       | 400798 | 140980 |
   5 | 5B   |        5 |  504 | Cluster 20           | 009504 1    |         9504 | Precinct 68     |  38.9367 |  -76.9908 |           | 2024-02-07 20:35:00
 | 2024-02-07 20:40:00 | 606076282 |                     5 |
| 397894 | 139390 | 24070589 | 2024-05-11 08:39:50 | DAY      | OTHERS | THEFT F/AUTO        | 2300 - 2599 BLOCK OF 9TH STREET NW        | 397894 | 139390 |
   1 | 1E   |        3 |  304 | Cluster 2            | 003500 2    |         3500 | Precinct 37     |  38.9224 |  -77.0243 |           | 2024-05-11 02:00:00
 | 2024-05-11 07:00:00 | 606076288 |                   300 |
+--------+--------+----------+---------------------+----------+--------+---------------------+-------------------------------------------+--------+--------+-
```

SELECT statement to display the first 10 rows.

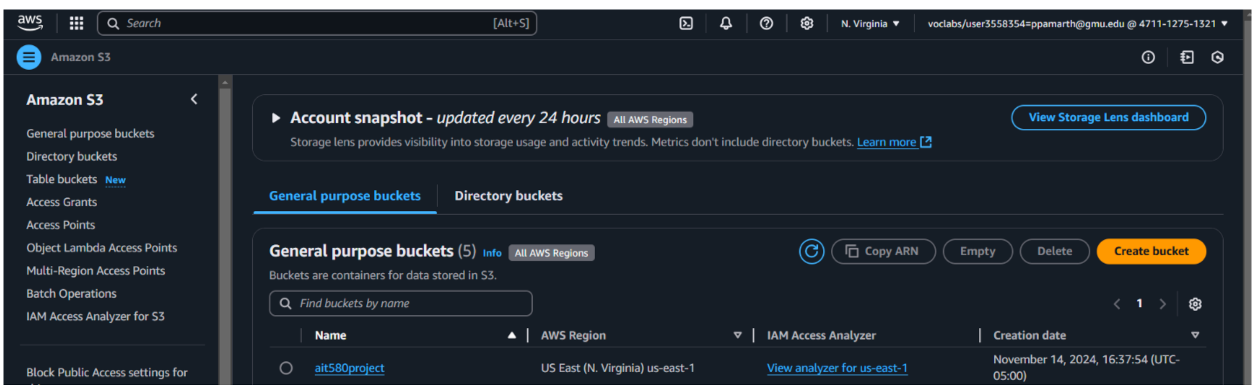
```
MySQL 9.0 Command Line Cli

   5 | 5B   |        5 |  504 | Cluster 20           | 009504 1    |         9504 | Precinct 68     |  38.9367 |  -76.9908 |           | 2024-02-07 20:35:00
 | 2024-02-07 20:40:00 | 606076282 |                     5 |
| 397894 | 139390 | 24070589 | 2024-05-11 08:39:50 | DAY      | OTHERS | THEFT F/AUTO        | 2300 - 2599 BLOCK OF 9TH STREET NW        | 397894 | 139390 |
   1 | 1E   |        3 |  304 | Cluster 2            | 003500 2    |         3500 | Precinct 37     |  38.9224 |  -77.0243 |           | 2024-05-11 02:00:00
 | 2024-05-11 07:00:00 | 606076288 |                   300 |
+--------+--------+----------+---------------------+----------+--------+---------------------+-------------------------------------------+--------+--------+-
-----+------+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+-----------+--------------------
-+---------------------+-----------+----------------------+
10 rows in set (0.00 sec)

mysql> select * from crime_data where OFFENSE='ROBBERY' limit 5;
+--------+--------+----------+---------------------+----------+--------+---------+-------------------------------------------+--------+--------+------+-----
-+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+------+---------------------+--------------
-------+-----------+-----------------------+
| X      | Y      | CCN      | REPORT_DAT          | SHIFT    | METHOD | OFFENSE | BLOCK                                     | XBLOCK | YBLOCK | WARD | ANC
 | DISTRICT | PSA  | NEIGHBORHOOD_CLUSTER | BLOCK_GROUP | CENSUS_TRACT | VOTING_PRECINCT | LATITUDE | LONGITUDE | BID  | START_DATE          | END_DATE
       | OBJECTID  | RESPONSE_TIME_MINUTES |
+--------+--------+----------+---------------------+----------+--------+---------+-------------------------------------------+--------+--------+------+-----
-+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+------+---------------------+--------------
-------+-----------+-----------------------+
| 398477 | 137709 | 24112374 | 2024-07-22 19:39:54 | EVENING  | OTHERS | ROBBERY | 400 - 499 BLOCK OF N STREET NW            | 398477 | 137709 |    2 | 2G
 |        3 |  308 | Cluster 7            | 004802 1    |         4802 | Precinct 18     |  38.9072 |  -77.0176 |      | 2024-07-22 13:57:00 | 2024-07-22 14
:00:00 | 606076269 |                     3 |
| 406531 | 135344 | 24064752 | 2024-04-30 17:49:51 | EVENING  | GUN    | ROBBERY | 5300 - 5399 BLOCK OF B STREET SE          | 406531 | 135344 |    7 | 7E
 |        6 |  604 | Cluster 33           | 009905 1    |         9905 | Precinct 105    |  38.8859 |  -76.9247 |      | 2024-04-30 16:28:00 | 2024-04-30 17
:52:00 | 606076292 |                    84 |
| 400595 | 139372 | 24058722 | 2024-04-19 18:33:47 | EVENING  | GUN    | ROBBERY | 900 - 999 BLOCK OF RHODE ISLAND AVENUE NE | 400595 | 139372 |    5 | 5C
 |        5 |  505 | Cluster 22           | 009102 4    |         9102 | Precinct 72     |  38.9222 |  -76.9931 |      | 2024-04-19 16:30:00 | 2024-04-19 16
:40:00 | 606076330 |                    10 |
| 398010 | 137857 | 24063325 | 2024-04-28 04:05:42 | MIDNIGHT | GUN    | ROBBERY | 8TH STREET NW AND O STREET NW             | 398010 | 137857 |    2 | 2G
 |        3 |  307 | Cluster 7            | 004901 2    |         4901 | Precinct 18     |  38.9086 |  -77.0229 |      | 2024-04-28 02:19:00 | 2024-04-28 03
:57:00 | 606076333 |                    98 |
| 397045 | 139950 | 24125992 | 2024-08-16 23:23:17 | MIDNIGHT | GUN    | ROBBERY | 1400 - 1499 BLOCK OF COLUMBIA ROAD NW     | 397045 | 139950 |    1 | 1A
 |        3 |  302 | Cluster 2            | 002802 3    |         2802 | Precinct 36     |  38.9274 |  -77.0341 |      | 2024-08-15 22:45:00 | 2024-08-15 22
:50:00 | 606076364 |                     5 |
+--------+--------+----------+---------------------+----------+--------+---------+-------------------------------------------+--------+--------+------+-----
-+----------+-----+----------------------+-------------+--------------+-----------------+----------+-----------+------+---------------------+--------------
-------+-----------+-----------------------+
5 rows in set (0.00 sec)

mysql> |
```

Ran a SELECT query to filter out to show only the first 5 rows of where OFFENCE is ROBBERY.

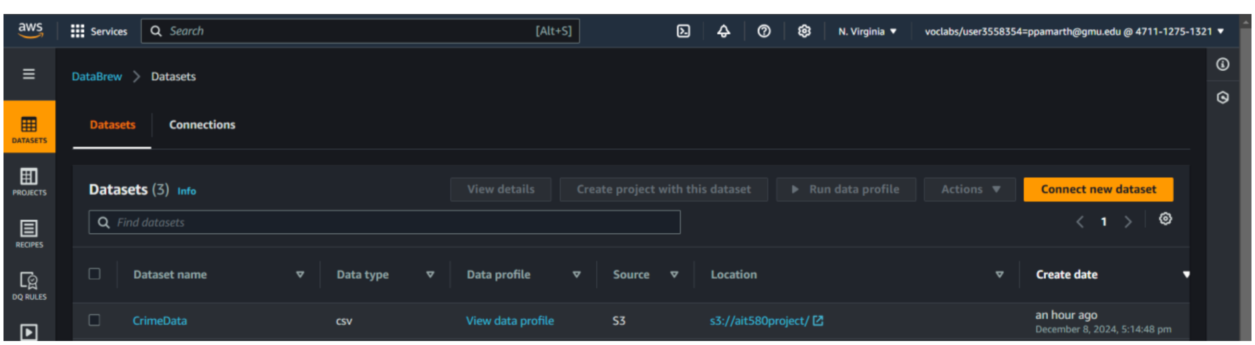
```
mysql> select OFFENSE, count(*) as offense_count
    -> from crime_data group by OFFENSE
    -> order by offense_count desc
    -> limit 5;
+---------------------+---------------+
| OFFENSE             | offense_count |
+---------------------+---------------+
| THEFT/OTHER         |         18734 |
| THEFT F/AUTO        |          9752 |
| MOTOR VEHICLE THEFT |          7176 |
| ROBBERY             |          2880 |
| BURGLARY            |          1372 |
+---------------------+---------------+
5 rows in set (0.06 sec)

mysql> 
```

In the SELECT statement I have taken the offence count which is grouped by offence type.

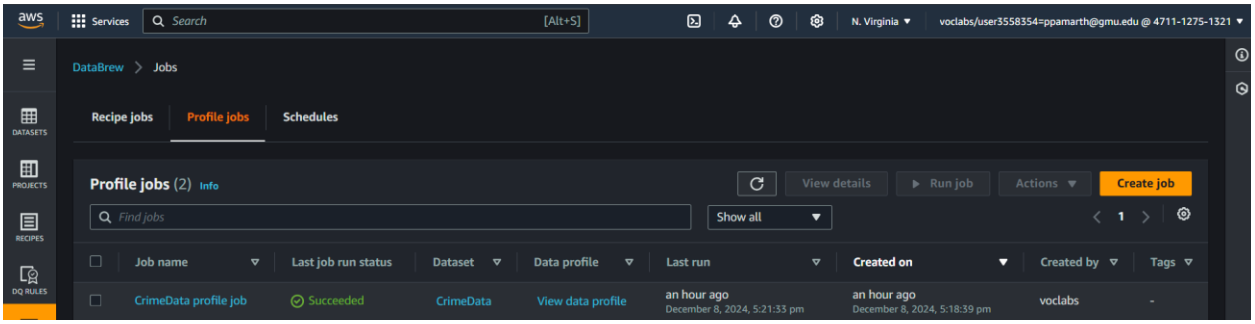
AWS Analysis



Created S3 bucket: - Here I created S3 bucket, loaded the dataset into the bucket.

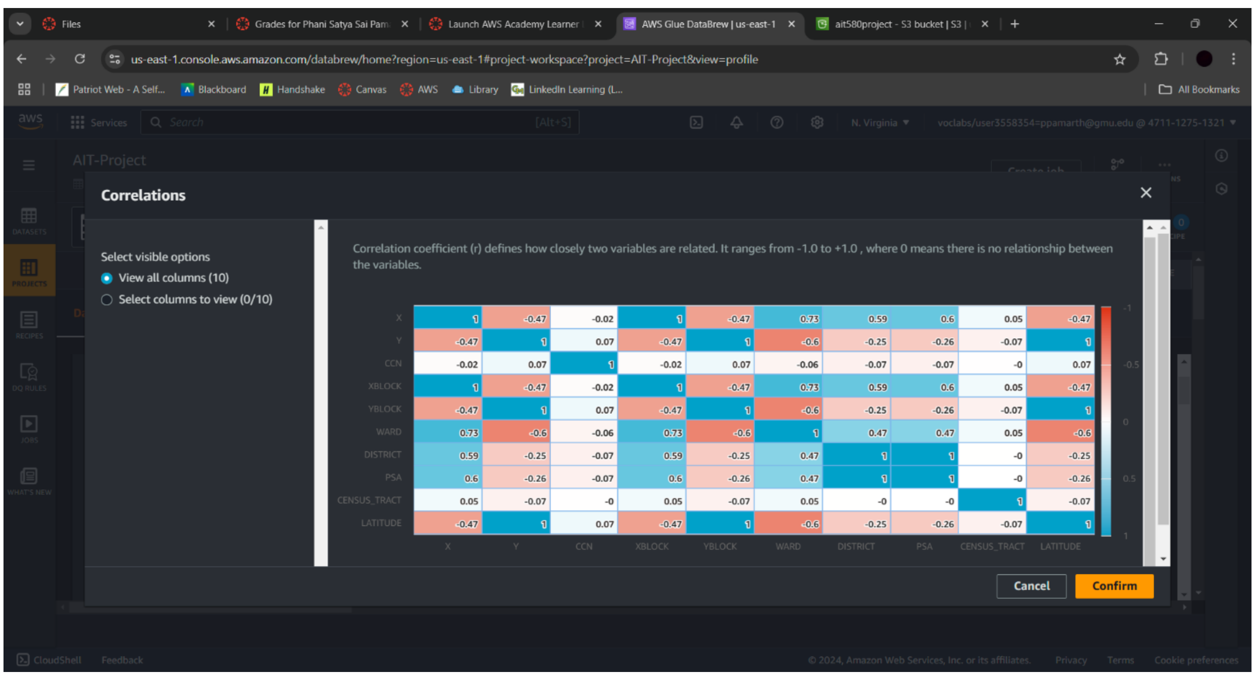


AWS Glue Data brew: - Here I loaded the dataset into Glue Databrew for data analysis.



For data analysis I created and ran the job in Glue databrew.

Here is the Data analysis in AWS S3 Bucket:



	X	Y	CCN	XBLOCK	YBLOCK	WARD	DISTRICT	PSA	CENSUS_TRACT	LATITUDE
X	1	-0.47	-0.02	1	-0.47	0.73	0.59	0.6	0.05	-0.47
Y	-0.47	1	0.07	-0.47	1	-0.6	-0.25	-0.26	-0.07	1
CCN	-0.02	0.07	1	-0.02	0.07	-0.06	-0.07	-0.07	-0	0.07
XBLOCK	1	-0.47	-0.02	1	-0.47	0.73	0.59	0.6	0.05	-0.47
YBLOCK	-0.47	1	0.07	-0.47	1	-0.6	-0.25	-0.26	-0.07	1
WARD	0.73	-0.6	-0.06	0.73	-0.6	1	0.47	0.47	0.05	-0.6
DISTRICT	0.59	-0.25	-0.07	0.59	-0.25	0.47	1	1	-0	-0.25
PSA	0.6	-0.26	-0.07	0.6	-0.26	0.47	1	1	-0	-0.26
CENSUS_TRACT	0.05	-0.07	-0	0.05	-0.07	0.05	-0	-0	1	-0.07
LATITUDE	-0.47	1	0.07	-0.47	1	-0.6	-0.25	-0.26	-0.07	1

The correlation matrix shows the relationship between the variables involved in the dataset. The correlation between WARD and DISTRICT is positive at 0.73, indicating they could be representative of similar geographic regions. X and CCN decrease as one increases due to their negative relationship of -0.47. Some of the very poor correlations such as those between XBLOCK and PSA, which have a correlation of -0.06 show little or no relationship between those variables. These will then help identify which factors are highly correlated and those that are not and may perhaps require less research.

Implications

These findings have practical applications to public safety and resource planning. Based on understanding response time, variation across districts and wards would enable the police department to better access routes of patrol and personnel to maximize the efficiency of response based on geographical area. Furthermore, the crime patterns observed in most landmarks underpin added surveillance and crime prevention measures necessary at the White House. Understanding how different crime methods link up with offense types and regions allows informed law enforcement strategy about the area.

Discussion and Conclusion

These findings highlighted critical trends and patterns necessary to shape workable law enforcement strategies as well as urban planning policies.

1) How do response times for crimes vary across districts and wards?

The investigation found significant differences in the average response times among districts and wards. Districts with slower responses can have issues related to inefficient infrastructure or the allocation of resources. These discrepancies in fact suggest that to ensure equal service delivery within the city, certain regions may require greater investments in emergency response capacity. Law enforcement will be able to gain community trust and increase public safety by helping to reduce these inequities.

2) Is proximity to certain landmarks associated with crime incidents?

These results reflected that crime rates were higher around all famous locations, such as the White House and Lincoln Memorial, which could be justified with a higher population, tourism, and commercial activities. Therefore, law enforcement organizations should invest resources to prevent crime in these areas and ensure safety for all citizens and tourists. However, to give a fuller knowledge of the dynamics of landmarks related to crimes, the research should in the future be complemented by added variables such as time of day and special events.

3) How do methods of crime vary with offense types and geographical regions?

The research showed that the method of crime varies with the incident type and location. For example, violent crimes more often involved firearms, while theft-related offenses did not, most of them were unarmed or opportunistic. Some of the localized issues in crime techniques are very prevalent in certain wards. These findings show how important region-specific crime prevention tactics are in empowering law enforcement to successfully combat distinctive patterns.

Conclusion

This study forms a base for an understanding of crime dynamics in Washington DC: regional differences in response times, landmark proximity and the effect of the same on the frequency of crimes, methodological differences, among others. The results highlight the specific actions law enforcement can take to better their response times within poor communities, prioritize safety around major landmarks, and implement regionally tailored crime prevention plans. These findings point out the utility of data-informed decision-making in metropolitan crime control in addition to providing a detailed blueprint for improved public safety.

The study also illustrates the ways in which advanced analytics and visualization techniques, such as those utilized in this work, can uncover unseen trends in criminal data. This approach will also enable lawmakers and law enforcement to manage newly arising threats and resources with greater ease. However, research is a continuous and time taking process due to the growing of urban environment at fast pace, leading to new ways to commit crimes and changes in crime trends. Thus, considering more variables including socioeconomic indicators, demographic shifts, and perhaps real-time data, this study could be expanded to give a better understanding of crime patterns in Washington, DC. Active research participants working together makes for a safer and strong community.

References

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